Team Details

a. **Team name: Syntax Squad**
b. **Team leader name: Sahasra Kona**
c. **Problem Statement:** Campus Safety and Emergency Response for Hyderabad Colleges

Brief about your solution and problem statement addressing

Large college campuses in and around Hyderabad—especially semi-urban zones—pose safety challenges for B.Tech students, particularly during late hours. Poor lighting, harassment risks, and sudden medical emergencies are often worsened by delayed response times and scattered emergency contact systems. Students currently lack a centralized, fast, and location-aware safety solution tailored to campus layouts.

Key Features:-

One-Tap SOS Button: Instantly sends emergency alerts with live location.

Real-Time Location Sharing: Uses GPS-based tracking for faster assistance.

Campus Safe Zone Map: Displays well-lit areas, security offices, and medical centers.

Voice-Based Emergency Reporting: Hands-free reporting using voice commands.

Safety Tips & Guidelines: Context-aware safety advice during emergencies.

Opportunities

a. How different is it from any of the other existing ideas?
b. How will it be able to solve the problem?

A)High-impact adoption in colleges: Hyderabad has a dense concentration of engineering colleges with large campuses, creating strong demand for a dedicated safety platform.
Women safety focus: Aligns with national and state-level women safety initiatives and college safety mandates.

B)Key Differentiators:-
Campus-first approach rather than general-purpose safety apps
Pre-mapped campus infrastructure (security rooms, medical centers, hostels)
Faster response due to direct alert routing to on-campus teams
Privacy-respecting design with controlled data sharing

List of features offered by the solution

1.One-Tap SOS Button A large, highly visible SOS button that instantly sends an emergency alert with no confirmation steps. Designed to work even in panic situations, ensuring speed and simplicity.

2. Live Location Sharing (Real-Time)Continuously tracks and updates the user's location using Google Maps, with visible GPS accuracy. Enables responders to reach the user without delays or confusion.

3. Emergency Alert System SOS alerts are sent simultaneously to:
Campus security dashboard Trusted contacts (friends/parents)Each alert includes user name, real-time location, time, and emergency type, creating a complete and reliable emergency flow.

4.Incident Heatmap (Analytics)Visualizes frequent alert locations to help colleges improve lighting, patrol routes, and infrastructure—showing long-term impact.

5.Security Dashboard (Web View)A centralized web interface for security teams showing:All active SOS alerts Live map with blinking alert markers Alert status: Active / Resolved This demonstrates system-level thinking beyond just a mobile app.

6.Safe Zones on Campus MapInteractive campus map highlighting : Security offices Hostels Medical centers Well-lit areas Users can tap and navigate to the nearest safe zone, enabling preventive safety. WOW Features (High-Impact Add-Ons)

7. Smart SOS Activation (Fake Alert Prevention)SOS triggers only via long press or multiple taps, reducing accidental alerts and improving system reliability.

8. Quick Emergency Type Selection Allows users to tag the emergency (medical, harassment, unsafe, etc.), helping security respond faster and more effectively.

9. Silent Mode SOS Triggers alerts silently without sound or vibration, ideal for high-risk situations where discretion is critical.

10. AI-Based Safety Tips (Lightweight)Provides real-time guidance like moving toward safe zones or staying in well-lit areas—smart, helpful, and demo-friendly.

Google Technologies used in the solution

- Google Maps Platform:-

Enables real-time GPS location tracking during emergencies
Displays campus safe zones and provides navigation
Helps responders quickly locate students on large campuses

- Firebase:-

Handles secure user authentication (fast sign-in)
Sends SOS alerts instantly in real time
Continuously updates live location to security dashboard
Push notifications to campus security and trusted contacts

- Dialogflow:-

Supports voice-based SOS activation
Allows hands-free emergency reporting
Useful during panic or medical emergencies

- Google Cloud (Optional / Extension):-

Manages backend logic using cloud functions
Supports analytics and incident heatmaps for long-term safety planning

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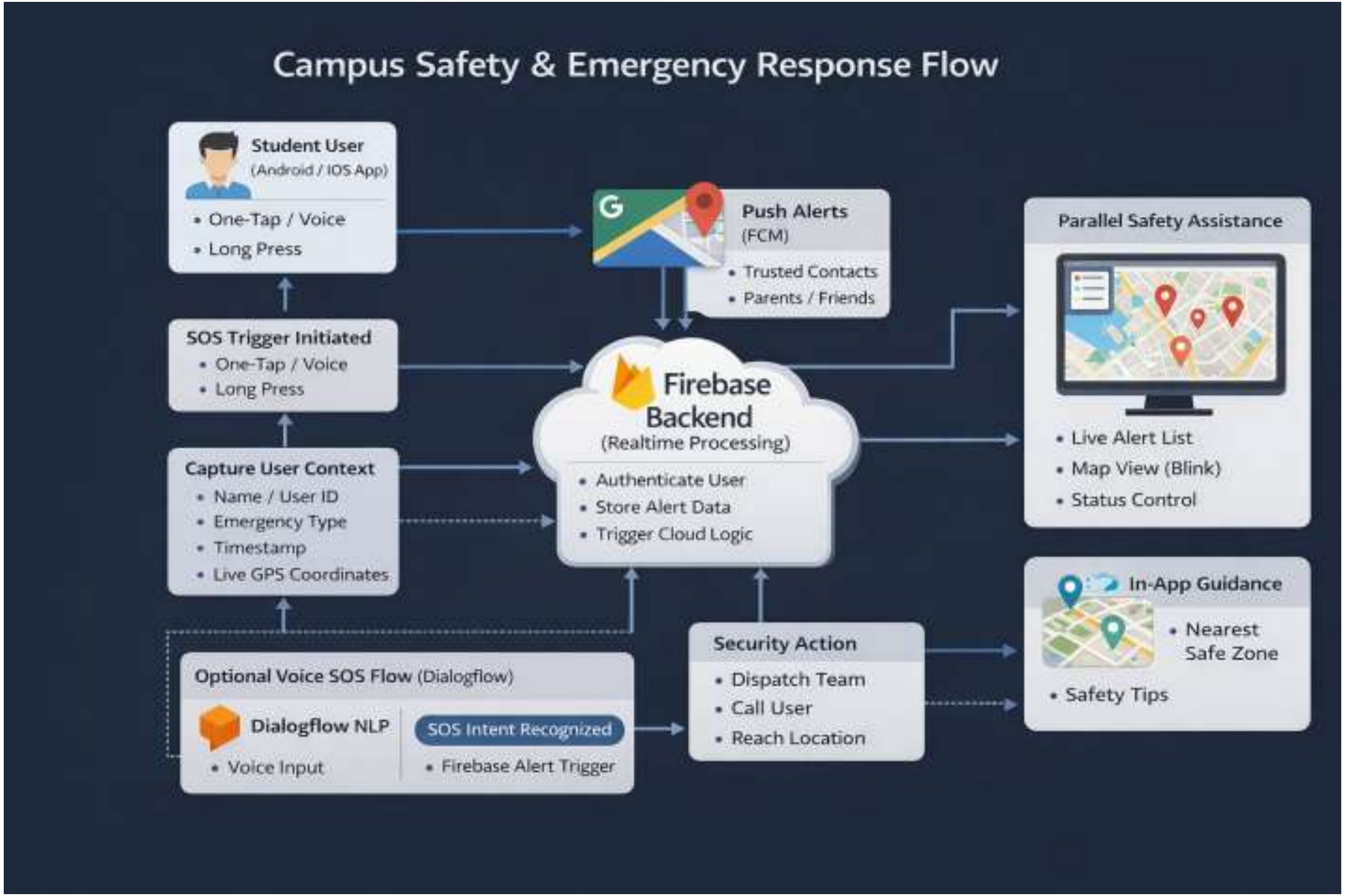
Process flow diagram or Use-case diagram



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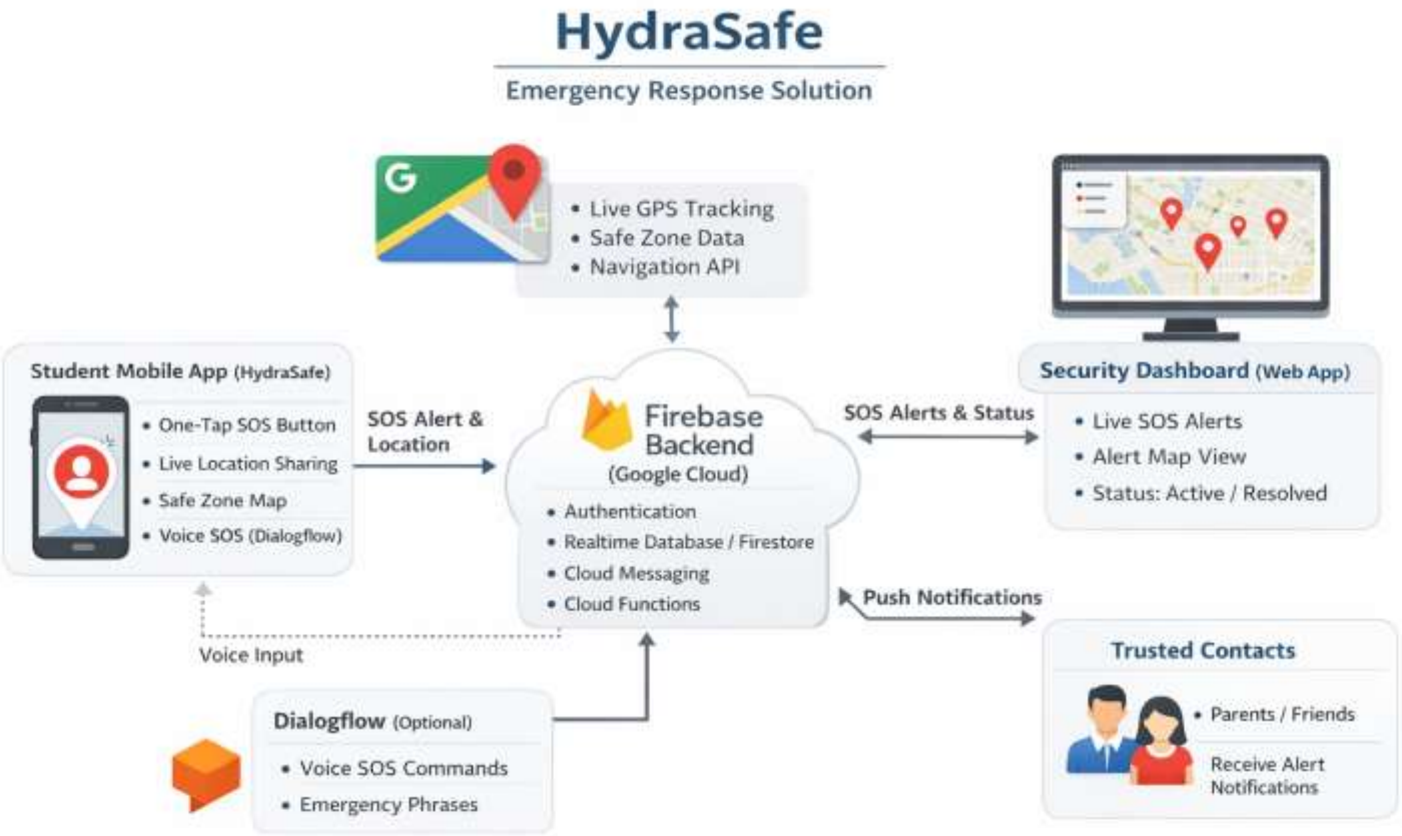
Wireframes/Mock diagrams of the proposed solution

Architecture diagram of the proposed solution



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Snapshots of the MVP

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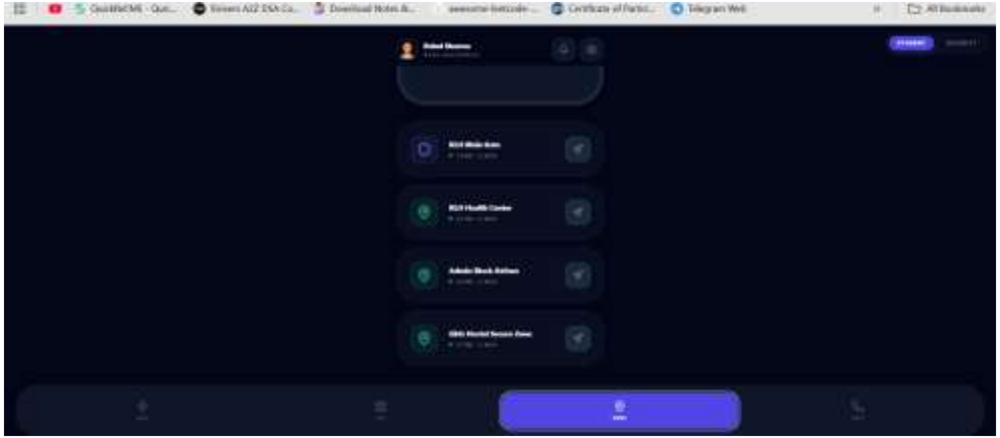

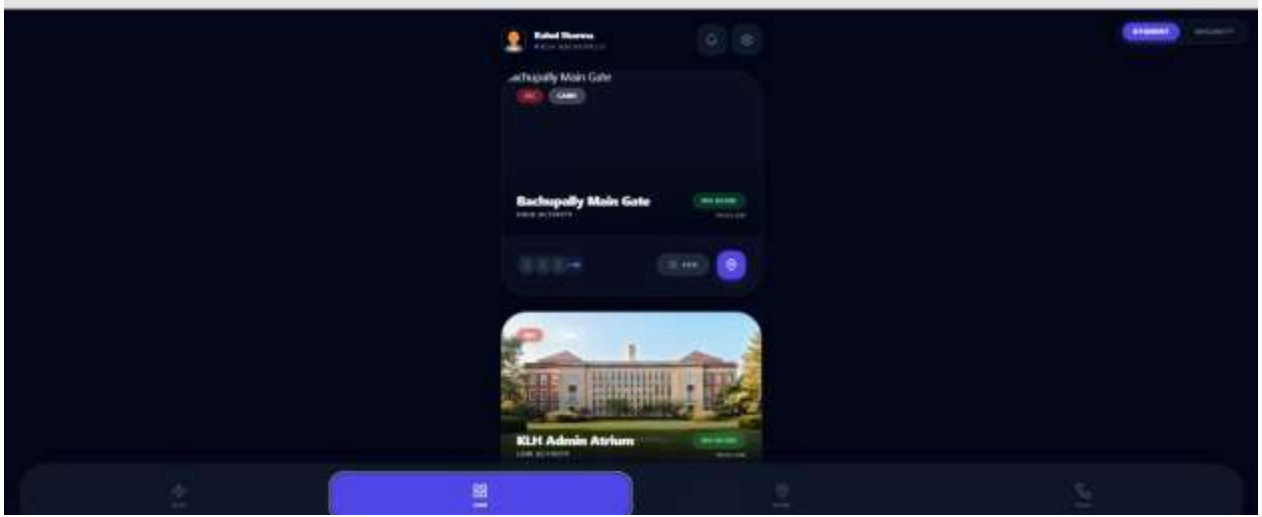

Additional Details/Future Development (if any)

- AI-Driven Incident Prediction:-

Use historical SOS data to predict high-risk zones and timings.
Smart alerts for students entering unsafe areas at late hours.

- Offline / Low-Network Mode:-

SMS-based SOS fallback when internet connectivity is weak.
Improves reliability in semi-urban or remote campus areas.

- Multi-Campus Expansion:-

Deploy the system across multiple colleges under a single admin panel.
Centralized monitoring for university groups or education boards.

- Integration with Local Authorities:-

Direct SOS escalation to local police stations, ambulances, and hospitals.
Faster real-world emergency response beyond campus boundaries.

Provide links to your:

1. GitHub Public Repository
[GitHub]

1. Demo Video Link (3 Minutes)
[Demo Video]

1. MVP Link
[App]

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Build Beyond Limits

Thank you!

Let's build beyond limits